

Balance-Field Framework (BFG): Canonical Reclosure and Structural Novelty

An Operator-Theoretic Inheritance-Emergence Theorem Derived from the Balance-Field Reclosure Equation

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22 September 2026

Scientific status

Pure mathematical preprint within the Balance-Field Framework (BFG). The results isolate and prove an operator-theoretic consequence of the canonical Balance-Field reclosure equation. They concern finite-dimensional self-adjoint operator reclosure under explicitly stated assumptions. No physical, biological, cognitive, or phenomenological realization is asserted.

Abstract

This paper extracts and formalizes an operator-theoretic inheritance-emergence theorem from the canonical reclosure equation of the Balance-Field Framework (BFG). Let Y be a nonnegative self-adjoint neutral load, let P be a persistent projection reducing Y , and let K be a self-adjoint formation operator. After restriction to the persistent support, the exact neutral pair $C=(I+Y)^{-1}$ and $B=Y(I+Y)^{-1}$ is combined with strictly positive reciprocal weights α and β into a two-channel analysis operator. Its polar isometry J canonically transports the active support, and the reclosed formation operator is $K_+=J^\dagger(K_P \oplus K_P)J$. The main theorem shows that K_+ is exactly a Schur multiplier of the inherited operator K_P in an eigenbasis of Y_P : $(K_+)_{ij}=\chi_{ij}(K_P)_{ij}$, where $\chi_{ij}=(\alpha+\beta y_i y_j)/\sqrt{(\alpha+\beta y_i^2)(\alpha+\beta y_j^2)}$. The multiplier matrix χ is a rank-at-most-two correlation matrix. Its fixed-point algebra is precisely the commutant of Y_P . Consequently, $[Y_P, K_P]=0$ if and only if $K_+=K_P$, while $[Y_P, K_P]\neq 0$ if and only if $\text{tr}(K_P^2)-\text{tr}(K_+^2)>0$, which forces a different spectrum. An exact weighted commutator identity quantifies the spectral change and yields two-sided norm bounds. If Y_P has at least two distinct eigenvalues, the no-novelty set is a proper linear subspace of the Hermitian operator space and therefore has Lebesgue measure zero. Repeated reclosure with fixed neutral geometry converges exponentially to spectral pinching onto the commutant of Y_P . An explicit two-dimensional family exhibits exact spectral reformation for every parameter $t>0$, while a separate counterexample proves that $[K, P]\neq 0$ alone is not sufficient for novelty. A 20,000-instance complex random audit, 5,000 commuting controls, and 5,000 ambient-support tests verify the closed formulas to floating-point precision. The contribution is not a new polar decomposition or Schur product theorem; it is the exact inheritance-novelty criterion induced by their specific canonical reclosure composition.

Keywords: Balance-Field Framework; BFG; Balance-Field reclosure equation; operator theory; polar decomposition; Schur multiplier; commutator; spectral novelty; recursive reclosure; inherited support; matrix analysis; emergence.

MSC 2020: 47A05, 47A30, 15A18, 15A45.

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1. Balance-Field origin, scope, and principal result

The present operator problem is the mathematically isolated reclosure problem arising inside the Balance-Field Framework (BFG): when a formed closure is neutrally split, restricted to its persistent support, transported canonically, and reclosed, does the successor inherit the same formation structure or acquire a genuinely new invariant formation spectrum? Recursive mathematical formalisms often use projection, normalization, or coarse-graining to define a successor state. That alone does not distinguish genuine structural change from the mere inheritance of a reduced support. The problem studied here is therefore precise: given a self-adjoint formation operator K , a nonnegative neutral load Y , and a persistent support P , when does the canonical BFG reclosure operator produce a successor formation operator whose invariant spectrum differs from the canonically inherited operator on that support?

The construction is the pure mathematical sector of the BFG Canonical Universal Reclosure Architecture [1]. In BFG terminology, Y is the positive neutral load generated by the closure geometry; $C=(I+Y)^{-1}$ and $B=Y(I+Y)^{-1}$ are the exact neutral retained/complementary pair; P is the persistent-support projector; the positive channel weights are supplied by endogenous reciprocal balance; and the polar isometry gives the canonical next-support transport. All physical, biological, cognitive, and phenomenological readings are deliberately removed here. What remains is a self-contained operator-theoretic question. Under the neutral-compatibility condition $[P,Y]=0$, the BFG polar-transport reclosure becomes a Schur multiplier on the compressed formation operator, yielding a necessary-and-sufficient commutator criterion for strict spectral novelty.

Main Theorem — Reclosure Inheritance-Emergence Dichotomy. Let H be a finite-dimensional complex Hilbert space, $Y=Y^\dagger \geq 0$, $K=K^\dagger$, and let $P=P^\dagger=P^2$ satisfy $[P,Y]=0$. Write $H_P=\text{Ran}(P)$, $Y_P=PYP|_{H_P}$, and $K_P=PKP|_{H_P}$. Let $\alpha, \beta > 0$ with $\alpha + \beta = 1$. Construct the exact neutral pair $C=(I+Y_P)^{-1}$, $B=Y_P(I+Y_P)^{-1}$, the two-channel analysis operator $A=[\sqrt{\alpha} C; \sqrt{\beta} B]$, its positive modulus $Q=(A^\dagger A)^{1/2}$,

polar isometry $J=AQ^{-1}$, and $K_+=J^\dagger(K_P \oplus K_P)J$. Then, in any eigenbasis $Y_P e_i = y_i e_i$, $K_+ = X \circ K_P$ with $\chi_{ij} = (\alpha + \beta y_i y_j) / \sqrt{((\alpha + \beta y_i^2)(\alpha + \beta y_j^2))}$. Moreover: (i) $K_+ = K_P$ iff $[Y_P, K_P] = 0$; (ii) if $[Y_P, K_P] \neq 0$ then $\text{tr}(K_P^2) > \text{tr}(K_+^2)$, hence $\sigma(K_+) \neq \sigma(K_P)$; (iii) the exact difference equals a positive weighted commutator sum. Thus strict formation-spectrum novelty occurs exactly when the inherited formation operator couples distinct neutral-load eigenspaces.

This theorem is stronger than a sufficient noncommutativity heuristic: within the stated stratum, it gives an if-and-only-if criterion. It also identifies which noncommutativity matters. The ambient commutator $[K, P]$ is neither necessary nor sufficient by itself; the relevant object is the commutator of the inherited operator K_P with the inherited neutral geometry Y_P .

2. Canonical reclosure on a neutral-compatible persistent support

BFG identification. The following objects are the mathematical reduction of the Balance-Field reclosure equation on one persistent sector: Y is the positive neutral load, P is the persistence projector, K is the formation operator, C and B are the exact neutral pair, and J is the canonical polar transport. Theorems below concern this BFG-derived operator core only.

2.1 Data and restriction

Let H be finite dimensional. Assume

$$Y=Y^\dagger \geq 0, \quad K=K^\dagger, \quad P=P^\dagger=P^2, \quad [P, Y]=0. \quad (1)$$

The persistent Hilbert space and inherited operators are

$$H_P := \text{Ran}(P), \quad Y_P := PYP|_{H_P}, \quad K_P := PKP|_{H_P}. \quad (2)$$

The condition $[P, Y]=0$ means that persistence and neutral geometry are compatible: H_P is a reducing subspace for Y . No assumption $[K, P]=0$ is made. Components of K coupling H_P to its orthogonal complement are discarded by canonical inheritance; novelty is assessed only after this inherited-support reduction.

2.2 Exact neutral pair

$$C := (I + Y_P)^{-1}, \quad B := Y_P(I + Y_P)^{-1}. \quad (3)$$

$$C+B=I, \quad 0 < C \leq I, \quad 0 \leq B < I, \quad [C, B] = [C, Y_P] = [B, Y_P] = 0. \quad (4)$$

Let $\alpha, \beta > 0$ with $\alpha + \beta = 1$. In the parent reclosure architecture these can be supplied by reciprocal load balance; the theorem itself needs only strict positivity.

2.3 Two-channel analysis and polar transport

$$A := [\sqrt{\alpha} C; \sqrt{\beta} B] : H_P \rightarrow H_P \oplus H_P. \quad (5)$$

$$Q := (A^\dagger A)^{1/2}, \quad J := AQ^{-1}. \quad (6)$$

Because $\alpha > 0$ and C is strictly positive, $A^\dagger A$ is strictly positive on H_P ; hence Q^{-1} exists. The map J is the unique polar isometry associated with A on the active support.

$$J^\dagger J = I_{H_P}. \quad (7)$$

$$K_+ := J^\dagger(K_P \oplus K_P)J. \quad (8)$$

Equation (7) is the reclosed formation operator. The inherited reference is K_P itself, not the full ambient K .

3. Exact polar reduction

Introduce the positive operator

$$R := (\alpha I + \beta Y_P^2)^{1/2}. \quad (9)$$

Since $C = (I + Y_P)^{-1}$ and $B = Y_P C$, one obtains

$$A^\dagger A = \alpha C^2 + \beta B^2 = (I + Y_P)^{-2} (\alpha I + \beta Y_P^2). \quad (10)$$

$$Q = (I + Y_P)^{-1} R. \quad (11)$$

$$J = [\sqrt{\alpha} R^{-1}; \sqrt{\beta} Y_P R^{-1}]. \quad (12)$$

Theorem 1 — Exact Polar Reclosure Formula. Under the assumptions of Section 2, the reclosed operator is

$$K_+ = R^{-1}(\alpha K_P + \beta Y_P K_P Y_P) R^{-1}. \quad (13)$$

Proof. Substitute the explicit polar isometry from Equation (11) into $K_+ = J^\dagger(K_P \oplus K_P)J$. The two diagonal channel contributions give $\alpha R^{-1} K_P R^{-1}$ and $\beta R^{-1} Y_P K_P Y_P R^{-1}$; factoring R^{-1} on both sides gives Equation (12).

The formula is basis-free. It already shows that reclosure is neither arbitrary conjugation nor simple compression: the neutral geometry appears on both sides of K_P inside a canonically normalized two-channel combination.

4. Schur representation and inheritance algebra

Let $\{e_i\}_{i=1}^r$ be an orthonormal eigenbasis of Y_P :

$$Y_P e_i = y_i e_i, \quad y_i \geq 0. \quad (14)$$

Define

$$\chi_{ij} := (\alpha + \beta y_i y_j) / \sqrt{(\alpha + \beta y_i^2)(\alpha + \beta y_j^2)}. \quad (15)$$

Theorem 2 — Schur Representation. In the eigenbasis of Y_P , canonical polar reclosure acts entrywise:

$$(K_+)_{ij} = \chi_{ij} (K_P)_{ij}, \quad \text{equivalently} \quad K_+ = X \circ K_P. \quad (16)$$

Proof. Equation (12) is diagonal on both sides in the Y_P eigenbasis. Its (i,j) entry is exactly the scalar in Equation (14) times $(K_P)_{ij}$.

4.1 The multiplier is a correlation matrix

Set

$$v_i := (\sqrt{\alpha}, \sqrt{\beta} y_i)^T / \sqrt{(\alpha + \beta y_i^2)} \in \mathbb{R}^2. \quad (17)$$

$$\chi_{ij} = \langle v_i, v_j \rangle. \quad (18)$$

Hence $X = (\chi_{ij})$ is positive semidefinite, has diagonal $\chi_{ii} = 1$, and rank at most two. By the Schur product theorem, $T_X: M_r(\mathbb{C}) \rightarrow M_r(\mathbb{C})$, $T_X(M) = X \circ M$, is a positive unital Schur multiplier; in finite dimensions a correlation-matrix Schur multiplier is completely positive. The use of a Schur multiplier here is a consequence of the polar reclosure, not an independent postulate.

$$1 - \chi_{ij}^2 = \alpha \beta (y_i - y_j)^2 / ((\alpha + \beta y_i^2)(\alpha + \beta y_j^2)). \quad (19)$$

Because $\alpha \beta > 0$, $\chi_{ij} = 1$ exactly when $y_i = y_j$.

4.2 Fixed-point algebra

Theorem 3 — Exact Inheritance Algebra. For any matrix M on H_P , $T_X(M) = M$ if and only if $[Y_P, M] = 0$. Therefore $\text{Fix}(T_X) = \{Y_P\}'$, the commutant of the inherited neutral load.

Proof. In the Y_P eigenbasis, $[Y_P, M]_{ij} = (y_i - y_j) M_{ij}$. By Equation (18), $\chi_{ij} = 1$ precisely on equal-eigenvalue blocks. Hence $X \circ M = M$ exactly when M_{ij} vanishes across unequal Y_P eigenspaces, which is equivalent to $[Y_P, M] = 0$.

5. Strict spectral novelty and the commutator identity

For Hermitian K_P , equality or inequality of formation spectra can be tested by unitary invariants. The first trace moment is preserved because $\chi_{ii} = 1$:

$$\text{tr}(K_+) = \text{tr}(K_P). \quad (20)$$

The second moment obeys an exact positive identity.

Theorem 4 — Exact Novelty Identity. For $K_P = K_P^\dagger$,

$$\text{tr}(K_P^2) - \text{tr}(K_+^2) = \alpha\beta \sum_{i,j} ((y_i - y_j)^2 |(K_P)_{ij}|^2) / ((\alpha + \beta y_i^2)(\alpha + \beta y_j^2)) \geq 0. \quad (21)$$

Proof. For Hermitian K_P , $\text{tr}(K_P^2) = \sum_{i,j} |(K_P)_{ij}|^2$ and $\text{tr}(K_+^2) = \sum_{i,j} \chi_{ij}^2 |(K_P)_{ij}|^2$. Subtract and insert Equation (18).

$$\| [Y_P, K_P] \|_F^2 = \sum_{i,j} (y_i - y_j)^2 |(K_P)_{ij}|^2. \quad (22)$$

Theorem 5 — Strict Spectral Novelty Criterion. For self-adjoint K_P , the following are equivalent: (a) $[Y_P, K_P] = 0$; (b) $K_+ = K_P$; (c) $\text{tr}(K_P^2) = \text{tr}(K_+^2)$; (d) $\sigma(K_+) = \sigma(K_P)$. Equivalently, $[Y_P, K_P] = 0$ if and only if $\sigma(K_+) = \sigma(K_P)$.

Proof. (a) $\Leftrightarrow$ (b) is Theorem 3. Equation (20) shows that equality of second moments holds exactly when every term with $y_i \neq y_j$ vanishes, i.e. exactly when $[Y_P, K_P] = 0$. If $[Y_P, K_P] = 0$, Equation (20) gives a strict reduction of $\text{tr}(K^2)$, so the eigenvalue multisets cannot agree. Conversely, $K_+ = K_P$ trivially implies equal spectra.

This is the inheritance–emergence dichotomy at the formation-spectrum level. It is stronger than saying that noncommutativity correlates with novelty: on this stratum, the relevant noncommutativity is exactly equivalent to spectral novelty.

6. Quantitative bounds and genericity

Let

$$a_i := \alpha + \beta y_i^2, \quad a_{\min} := \min_i a_i, \quad a_{\max} := \max_i a_i. \quad (23)$$

Theorem 6 — Quantitative Commutator Control. The loss of the second spectral moment is bounded above and below by the Frobenius commutator norm:

$$(\alpha\beta/a_{\max}^2) \| [Y_P, K_P] \|_F^2 \leq \text{tr}(K_P^2) - \text{tr}(K_+^2) \leq (\alpha\beta/a_{\min}^2) \| [Y_P, K_P] \|_F^2. \quad (24)$$

Proof. In Equation (20), each denominator $a_i a_j$ lies between $a_{\min}^2$ and $a_{\max}^2$.

6.1 Generic novelty

Suppose Y_P has s distinct eigenvalues with multiplicities $m_1, \dots, m_s$, where $\sum m_a = r$. The real vector space $\text{Herm}(r)$ has dimension r^2 . Its commutant with Y_P consists of Hermitian block matrices on the eigenspaces and has real dimension $\sum_a m_a^2$.

$$\text{codim}_{\text{Herm}(r)} \{K_P : [Y_P, K_P] = 0\} = r^2 - \sum_a m_a^2 = 2 \sum_{a < b} m_a m_b. \quad (25)$$

Corollary 6.1 — Generic Strict Novelty. If Y_P is not a scalar multiple of the identity, the no-novelty set $\{K_P \in \text{Herm}(r) : [Y_P, K_P] = 0\}$ is a proper linear subspace of positive codimension and therefore has Lebesgue measure zero. Hence, for almost every Hermitian inherited formation operator K_P , canonical polar reclosure produces strict spectral novelty.

If Y_P is scalar, the opposite extreme holds: X is the all-ones matrix and $K_+ = K_P$ for every K_P . Neutral differentiation is therefore necessary for this novelty mechanism.

7. Iterated reclosure and convergence to pinching

When Y_P , α , and β are frozen and the same reclosure map is iterated, the Schur representation makes the long-time behavior exact.

$$T_X^n (K_P)_{ij} = \chi_{ij}^n (K_P)_{ij}. \quad (26)$$

Let $\{\Pi_\lambda\}$ be the spectral projections of Y_P and define the spectral pinching

$$E_Y(K_P) := \sum_{\lambda \in \sigma(Y_P)} \Pi_\lambda K_P \Pi_\lambda. \quad (27)$$

Theorem 7 — Convergence to the Inheritance Algebra. For fixed Y_P and $\alpha, \beta > 0$, $T_X^n(K_P) \rightarrow E_Y(K_P)$ in every matrix norm. If $\chi_* := \max\{\chi_{ij} : y_i \neq y_j\} < 1$, then

$$\|T_X^n(K_P) - E_Y(K_P)\|_F \leq \chi_*^n \|K_P - E_Y(K_P)\|_F. \quad (28)$$

Proof. Equal-eigenvalue blocks have $\chi_{ij} = 1$ and are fixed. Every cross-eigenspace entry has $0 < \chi_{ij} < 1$ and decays geometrically under iteration.

Thus frozen neutral geometry does not generate unbounded novelty indefinitely. It contracts cross-neutral couplings toward the commutant of Y_P . Further novelty across generations requires the neutral geometry itself, the persistent support, the reciprocal weights, or the reconstructed formation operator to change.

8. Formation gate and a rigorous emergence corollary

Spectral novelty alone does not guarantee a newly formed nontrivial branch. To connect the operator theorem to determinate formation, impose the same quartic formation gate used in the motivating reclosure architecture [1].

$$A_{\text{form},+}(\delta) := \frac{1}{2} \langle \delta, K_+ \delta \rangle + \frac{1}{4} \|\delta\|^4. \quad (29)$$

Let $\lambda_{\{+,0\}} = \min \sigma(K_+)$. Assume

$$\lambda_{\{+,0\}} < 0, \quad \text{mult}(\lambda_{\{+,0\}}) = 1, \quad \Delta_+ := \lambda_{\{+,1\}} - \lambda_{\{+,0\}} > 0. \quad (30)$$

$$\delta^*_{\{+,\pm\}} = \pm \sqrt{-\lambda_{\{+,0\}}} e_{\{+,0\}}. \quad (31)$$

Corollary 8.1 — Strict Formational Emergence. Assume the hypotheses of Theorem 5 and the formation gate in Equation (28). If $[Y_P, K_P] \neq 0$, then K_+ has a spectrum distinct from the inherited K_P and the quartic successor functional has exactly two nonzero polarity-related global minima. The successor is therefore both formed and spectrally non-inherited.

This corollary is deliberately narrower than a universal theory of emergence. It gives an exact sufficient-and-necessary novelty criterion for the formation spectrum within the neutral-compatible reclosure stratum, plus an independent gate for nontrivial formation.

9. Why projection noncommutativity alone is insufficient

A tempting criterion is $[K, P] \neq 0$. It is false as a sufficient condition for novelty.

Take

$$K = \text{diag}(-3, 1, 4), \quad Y = I_3. \quad (32)$$

Let P project onto $\text{span}\{u, e_3\}$, with $u = (e_1 + e_2)/\sqrt{2}$. Then

$$\|[K, P]\|_2 = 2 \neq 0. \quad (33)$$

In the orthonormal persistent basis $\{u, e_3\}$,

$$K_P = \text{diag}(-1, 4), \quad Y_P = I_2. \quad (34)$$

Because Y_P is scalar, Theorem 3 gives

$$K_+ = K_P. \quad (35)$$

The formation gate may still pass: K_P has a simple negative eigenvalue -1 with positive gap 5. Hence a successor formation can exist without strict novelty. The ambient commutator $[K, P]$ measures mismatch with the discarded complement, not novelty within the canonically inherited support.

10. Exact two-dimensional reformation family

A closed analytic family demonstrates strict novelty without numerical approximation. Set, for $t > 0$,

$$P = I_2, \quad Y(t) = \text{diag}(0, t), \quad K = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}, \quad D = (1, 1)^T. \quad (36)$$

With $G = I + Y$, the exact persistent neutral loads of the retained and complementary channels are

$$\Lambda_{\text{keep}}=(t+2)/(t+1), \quad \Lambda_{\text{up}}=t^2/(t+1). \quad (37)$$

Reciprocal balance gives

$$\alpha=\omega_{\text{keep}}=t^2/(t^2+t+2), \quad \beta=\omega_{\text{up}}=(t+2)/(t^2+t+2). \quad (38)$$

The off-diagonal reclosure factor is

$$\chi_{12}=1/\sqrt{t+3}. \quad (39)$$

$$K_{+}(t)=[[-2, 1/\sqrt{t+3}], [1/\sqrt{t+3}, 2]]. \quad (40)$$

The spectra are

$$\sigma(K)=-\sqrt{5}, \sqrt{5}. \quad (41)$$

$$\sigma(K_{+}(t))=-\sqrt{4+1/(t+3)}, \sqrt{4+1/(t+3)}. \quad (42)$$

For every $t>0$, $4+1/(t+3)<5$, so the two spectra are exactly different. Moreover $K_{+}(t)$ has a simple negative minimum eigenvalue for every $t>0$, so the formation gate is passed throughout the family.

Proposition 10.1 — Exact One-Parameter Emergence Family. For every $t>0$, the canonical reciprocal reclosure of Equations (33)–(38) is both spectrally novel and determinately formable. No floating-point calculation is required for this conclusion.

11. Numerical falsification audit

The analytical results make simulation unnecessary for proof, but numerical stress tests are useful for detecting algebraic or implementation errors. All tests used seed 20260922 and standard double-precision complex linear algebra.

Audit	Trials	Observed	Analytic expectation
Direct polar route vs. closed Schur formula	20,000	max relative Frobenius error 7.90×10^{-15}	Exact equality
Trace-loss identity, Eq. (20)	20,000	max absolute discrepancy 3.62×10^{-13}	0
Two-sided commutator bounds, Eq. (23)	20,000	0 violations	No violations
Strict novelty in continuously sampled non-scalar Y	20,000	20,000 strict positive second-moment losses	Almost sure
Commuting degenerate-Y controls	5,000	$\max \ K_{+} - K_P\ _F = 1.07 \times 10^{-14}$	0
Commuting controls: spectrum	5,000	max eigenvalue discrepancy 1.15×10^{-14}	0
Ambient P with $[P, Y]=0$ vs. restricted formula	5,000	max relative error 3.92×10^{-13}	Exact equality

The numerical audit attempted to falsify the closed formulas rather than to establish them. The largest errors occur in support-reconstruction cases with small singular values and remain consistent with floating-point conditioning. The theorem rests on the exact derivations in Sections 3–6, not on the trial count.

12. Relation to established operator theory and priority boundary

The present theorem uses established mathematics. Polar decomposition and its canonical partial isometry are standard tools in matrix and operator theory [2–4]. The Schur product theorem and correlation-matrix Schur multipliers are classical [2,3]. Spectral pinching and block-diagonal fixed algebras are standard [5]. Generalized dephasing semigroups similarly damp cross-eigenspace matrix elements and have invariant algebras determined by spectral blocks [6]. Commutator norm inequalities are a substantial existing literature [7,8].

The contribution claimed here is therefore not any of those ingredients in isolation. It is the exact reduction of this particular canonical two-channel polar reclosure to a correlation-matrix Schur multiplier, followed by the equivalence between neutral-load noncommutation and strict formation-spectrum novelty, together with the exact weighted commutator identity and its inherited-support interpretation.

Object	Literature status	Role here
Polar decomposition / partial isometry	Established	Used to define canonical support transport.
Schur product / correlation-matrix multiplier	Established	The reclosure multiplier X is derived, not postulated.
Spectral pinching / commutant fixed algebra	Established	Appears as the fixed-geometry iteration limit.
Commutator norm theory	Established	Provides context; Eq. (20) is a reclosure-specific exact identity.
Exact X generated by neutral pair + reciprocal two-channel polar normalization	Not located as an identical construction in the targeted search	Primary structural contribution.
$[Y_P, K_P]$ iff inheritance vs. strict spectral novelty for this reclosure	Not located as an identical theorem in the targeted search	Primary theorem-level contribution.

Priority caution. A targeted literature search performed through 22 September 2026 located close mathematical ingredients but no publication with the identical reclosure map and inheritance–novelty theorem. This is not an exhaustive or legal priority determination. Any submission should invite comparison with adjacent work on Schur multipliers, dephasing channels, operator means, conditional expectations, and noncommutative coarse-graining.

13. Scope, limitations, and extensions

- Finite-dimensional scope. The exact Schur theorem is stated for finite-dimensional H_P . Extension to unbounded operators requires domain control for Y , K , and the polar transport.
- Neutral-compatible persistence. The equivalence theorem assumes $[P, Y]=0$. Without this reduction property, the general defect identity remains available, but the simple Schur formula need not hold.
- Spectral novelty versus complete structural novelty. A changed formation spectrum is a strong invariant witness of novelty, but equality of formation spectra would not by itself prove complete equivalence of all other state data.
- Formation is a separate gate. $[Y_P, K_P]=0$ guarantees spectral change, not the existence of a simple negative eigenvalue. The formation gate must be checked independently.
- Frozen-geometry iteration converges to inheritance. Persistent multi-generation novelty therefore requires evolving Y , P , weights, or reconstructed capacities rather than repeated application of one fixed multiplier.
- Empirical interpretation is outside scope. The theorem is a statement about operators, not a demonstration that any natural system instantiates the reclosure architecture.

14. Conclusion

The canonical reclosure equation of the Balance–Field Framework admits an exact operator-theoretic inheritance–emergence criterion on a neutral-compatible persistent support. Once the support is fixed, the two neutral channels and their positive weights generate a rank-at-most-two correlation matrix X . The successor formation operator is exactly $K_+=X \circ K_P$ in the neutral-load eigenbasis. The fixed-point algebra of this map is the commutant of Y_P . Consequently, commuting inherited formation is exactly canonical inheritance, while noncommuting inherited formation necessarily changes the second spectral moment and therefore the spectrum.

The resulting statement is both qualitative and quantitative: the commutator $[Y_P, K_P]$ decides whether strict formation-spectrum novelty occurs, and its Frobenius norm controls the magnitude of the second-moment change through explicit two-sided bounds. If the neutral geometry has more than one eigenspace, inheritance is nongeneric in the space of Hermitian formation operators. Yet frozen repeated reclosure converges back toward the commutant, preventing an interpretation of recursion alone as inexhaustible novelty.

The mathematical conclusion is therefore narrow but exact: within the stated canonical reclosure stratum, genuine spectral reformation is neither a naming convention nor a numerical accident. It occurs exactly when the inherited formation operator contains couplings across distinct neutral-load eigenspaces and the canonical polar transport attenuates those couplings in a way that cannot preserve the spectrum.

Appendix A. Proof register

A.1 Correlation-matrix structure

For each neutral eigenvalue y_i define the normalized vector $v_i = (\sqrt{\alpha}, \sqrt{\beta} y_i)^T / \sqrt{(\alpha + \beta y_i^2)}$. Then $\chi_{ij} = v_i^T v_j$. Thus X is a Gram matrix, so $X \geq 0$ and $\chi_{ii} = 1$. Equality $\chi_{ij} = 1$ is equality in Cauchy–Schwarz. Because the first coordinate of every v_i is positive, equality implies $v_i = v_j$ and hence $y_i = y_j$.

A.2 Strictness of the second-moment identity

Every term in Equation (20) is nonnegative. Since $\alpha\beta > 0$ and the denominators are positive, the sum vanishes exactly when $(y_i - y_j)(K_P)_{ij} = 0$ for every i, j , i.e. when $[Y_P, K_P] = 0$. This proves strictness without perturbation theory.

A.3 Dimension of the no-novelty subspace

If Y_P has eigenspace dimensions $m_1, \dots, m_s$, then a Hermitian operator commuting with Y_P is an arbitrary Hermitian block on each eigenspace and has real dimension $\sum m_a^2$. The full Hermitian space has dimension r^2 . For $s \geq 2$, $r^2 - \sum m_a^2 = 2 \sum_{a < b} m_a m_b > 0$; therefore the commuting set is a proper linear subspace and has Lebesgue measure zero.

A.4 Iteration

Since Schur multiplication composes entrywise, $T_X^n(M) = X^{\wedge n} \circ M$. On equal- y blocks $\chi_{ij} = 1$; across distinct y blocks $0 < \chi_{ij} < 1$. The claimed pinching limit and geometric rate follow immediately.

A.5 General defect identity outside the neutral-compatible stratum

Without $[P, Y] = 0$ one may still define the ambient analysis operator $A = [\sqrt{\alpha} PC; \sqrt{\beta} PB]$, $Q = (A^\dagger A)^{1/2}$, and polar partial isometry $J = A Q^\dagger$. Let P_Q be the support projection of Q and define $K_{inh} = P_Q K P_Q$ on $\text{supp}(Q)$. Then

$$K_- - K_{inh} = Q^\dagger [A^\dagger (K \oplus K) A - Q K Q] Q^\dagger. \quad (43)$$

Equation (39) is exact wherever the products are defined. It provides a general reclosure-defect operator, but it does not by itself yield the if-and-only-if Schur criterion of the neutral-compatible stratum.

Appendix B. Reproducibility protocol

Item	Specification
Random seed	20260922
Primary random trials	20,000 complex Hermitian K ; dimensions 2–9
Neutral load	Random unitary conjugate of $\text{diag}(\exp(U[-2.5, 2.5]))$
Weights	α uniform on $[0.02, 0.98]$, $\beta = 1 - \alpha$
Direct route	$C = (I + Y)^{-1}$, $B = YC$, $A = [\sqrt{\alpha} C; \sqrt{\beta} B]$, $J = A(A^\dagger A)^{-1/2}$, $K_+ = J^\dagger (K \oplus K) J$
Closed route	Diagonalize Y and apply χ_{ij} entrywise
Commuting controls	5,000 tests with degenerate Y and K block-Hermitian inside Y eigenspaces
Ambient-support tests	5,000 tests with spectral P commuting with Y and arbitrary ambient K
Floating-point interpretation	Audit only; all theorems are analytical

Minimal pseudocode:

1. Draw $Y \in \mathbb{C}^{r \times r}$ and $K = K^\dagger$; choose $\alpha \in (0, 1)$, $\beta = 1 - \alpha$.
2. Compute $C = (I + Y)^{-1}$, $B = YC$, $A = [\sqrt{\alpha} C; \sqrt{\beta} B]$.
3. Compute $Q = (A^\dagger A)^{1/2}$ and $J = A Q^{-1}$.
4. Form $K_{direct} = J^\dagger (K \oplus K) J$.
5. Diagonalize $Y = U \text{diag}(y) U^\dagger$; set $\hat{K} = U^\dagger K U$.

6. Build $\chi_{ij}=(\alpha+\beta y_{iy_j})/\sqrt{((\alpha+\beta y_{i^2})(\alpha+\beta y_{j^2}))}$.
7. Form $K_{\text{closed}}=U(X\circ\text{Khat})U^\dagger$ and compare.
8. Compare $\text{tr}(K^2)-\text{tr}(K_{\text{direct}}^2)$ with Equation (20).

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Declarations

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Author contribution. Conceptualization, mathematical formulation, theorem construction, interpretation, and manuscript responsibility: Marcel Theodor Wende.

Data availability. No empirical dataset was analyzed. Numerical audits use synthetic random matrices and the reproducibility protocol in Appendix B.

Code availability. The algorithmic construction and random seed are specified in Appendix B; no external proprietary code is required.

Competing interests and funding. No external funding or competing-interest claim is made in this preprint; journal-specific declarations should be updated at submission.

Scope statement. This manuscript is the pure mathematical operator-theoretic sector of the Balance-Field Framework (BFG) reclosure architecture. It makes no empirical claim about natural systems.